University of Ottawa

Faculty of Engineering
School of Electrical Engineering and Computer Science

Assignment 2

Normalization and Functional Dependencies

Course CSI2132	Databases I
Academic year	2025-26
Section	A & B
Semester	Winter
Instructor	Verena Kantere
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Every student must submit the assignment individually

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Exercise 1 (10%)

Consider a relation schema R (A, B, C) and FDs the set F:

F= {
A → BC
B → C
A → B
AB → C
}

a) $F = \{A \to B, A \to C, B \to C, AB \to C\}$.

$\Rightarrow F = \{A \to B, A \to C, B \to C\}$.

$F_c = \{A \to B, B \to C\}$.

a. (7%) Find the canonical cover of F.
b. (3%) Find the minimal cover of F.

b): $F_{min} = \{A \to B, B \to C\}$.

Exercise 2 (20%)

Consider a relation schema R (A, B, C, D) and FDs the set F:

F= {
AB → C
C → D
D → A
}

a): {AB} & {BC}.
b): BCNF: C->D & D->A.

a. (7%) Find all the candidate keys of R.
b. (3%) Indicate all BCNF violations for R.
c. (7%) Decompose R in BCNF.
d. (3%) Indicate which dependencies in F, if any, are not preserved by the BCNF decomposition.

c) R1(C,D) with key {C} R2{A,B,C} with {AB} & {BC}

d: No, D->A.

Exercise 3 (20%)

Consider R = (A, B, C, D) with FDs the set F:

F = {
AB → C
B → D
C → A
}

a. (7%) Find all the candidate keys of R. {AB} & {BC}.
b. (3%) Indicate all BCNF violations for R. B->D, C->A
c. (7%) Decompose R in BCNF.
d. (3%) Indicate which dependencies in F, if any, are not preserved by the BCNF decomposition.

c): R(B,D) with key{B}, R(A,C) with key {C}. R(B,C) with {BC}

d.) AB->C is not

Exercise 4 (20%)

Consider a relation schema R (A, B, C, D, E) and FDs the set F:

F ={
B → EA
EBC → D
BED → A
}

{B->E, B->A, EBC->D. BED->A}

a. (5%) Find the candidate keys of R. {BC}
b. (5%) Find the canonical cover and the minimal cover of R.
c. (3%) Find the highest (i.e. strictest) normal form (no NF, 1NF, 2NF, 3NF, BCNF) that R satisfies. Explain your answer.
d. (7%) If R is not in 3NF, decompose it in 3NF.

b.) F = {B->E, B->A, BC->D, ~~B->A~~}.

c. 1NF

d. R(ABE) with {B}, R(B,C,D) with {BC}

Exercise 5 (30%)

Consider a relation schema R (A, B, C, D) and FDs the set F:

F ={

A→B

A→C

C→D

}

a. (3%) Find the candidate keys of R. {A}

b. (3%) Determine the highest normal form (no NF, 1NF, 2NF, 3NF, BCNF) that R satisfies. Explain your answer. 2NF

c. (7%) Decompose R so that the decomposition satisfies the next higher normal form than the one R itself satisfies. R(A,B,C) with {A} R(C,D) with {C}.

d. (5%) Show the canonical and minimal covers of functional dependencies for the decomposed relation schemas. {C→D}.

e. (5%) List the candidate keys of the decomposed relation schemas.

f. (7%) Prove that the decomposition is a lossless-join decomposition and/or dependency preserving.

e: $A^+ = \{A, B, C\} \Rightarrow \{A\}$

$C^+ = \{C, D\} \Rightarrow \{C\}$

by: $R(A,B,C) \Rightarrow A \to B,\ B \to C$

$R(C,D) \Rightarrow C \to D.$

∴ all FDs are preserved.

∴ lossless join.

$\underset{R_1}{R(A,B,C)} \cap \underset{R_2}{R(C,D)} = \{C\}.$

also: since: $R_2 \setminus R_1 = \{D\}$. ∴ $C \to D$ ✓

∴ it's lossless join.